

2.2

POUNDS

THE AMOUNT OF PLUTONIUM NEEDED FOR AN EXPLOSION EQUAL TO THAT PRODUCED BY **20,000 TONS OF CHEMICAL EXPLOSIVE**

The production method of the glowing radioactive element plutonium was initially kept secret because it was key to the manufacture of early atomic bombs.

IN THE FIRST PART OF THE 20TH CENTURY, geologists used data from studying **seismic waves** to conclude that Earth had a distinct core (see 1906). By studying the way waves were transmitted during earthquakes, they deduced that the core was made up of different materials from the rest of the planet. In 1940, Canadian geologist **Reginald Aldworth Daly** (1871–1957) published ***Strength and Structure of the Earth***, in which he identified a multi-layered arrangement around

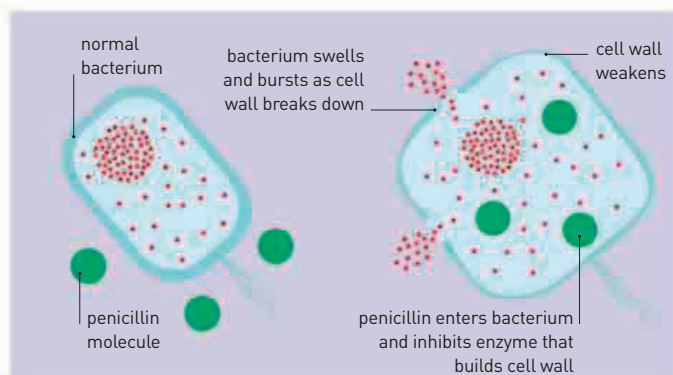
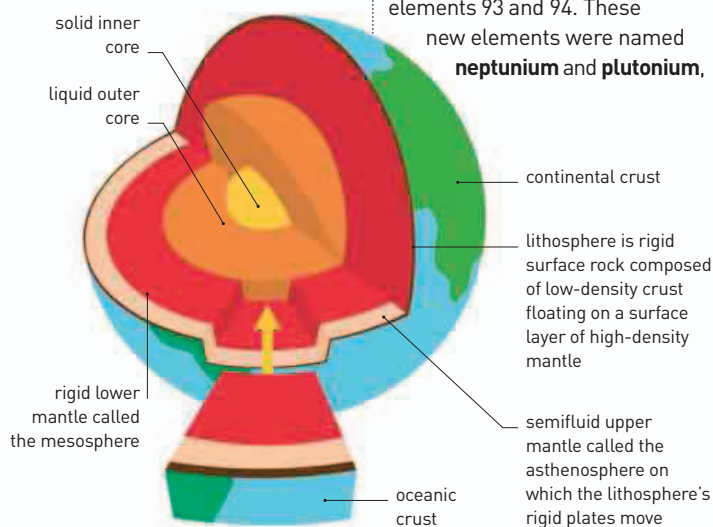
Layers of outer Earth

Earth's iron core is surrounded by a thick mantle covered in crust. The hard rock of the uppermost mantle and crust forms the movable plates that account for drifting continents.

Earth's core. He stated that a rigid but brittle envelope overlaid a much thicker semisolid layer, which was hotter toward the core. Today, it is now known that **Earth's layers differ chemically and physically**, with a silicon-rich rocky surround and a core that is roughly 80 percent iron (see below). The rigid surround includes a low-density surface crust, with a mantle layer made up of higher-density rock beneath it.

In 1940, at University of California, Berkeley, scientists successfully made the **first transuranium elements**—those with an atomic number higher than that of uranium (92). By using a particle accelerator, scientists successfully made elements 93 and 94. These

new elements were named **neptunium and plutonium**,



HOW PENICILLIN WORKS

Penicillin is an antibiotic—a member of a group of chemicals that either prevent the growth of bacteria or kill them altogether. These chemicals attack targets that are present only in bacterial cells and not in infected tissue. Penicillin inhibits the process that many bacteria use to build their cell walls. As a result, the cell walls weaken, making the bacteria absorb water and burst.

respectively, after Neptune and Pluto—the two outer planets of the Solar System. Publication of the discovery of these elements was delayed until after World War II as it was found that plutonium had the potential to be used as fuel for an atomic bomb.

More than 30 years after Scottish bacteriologist Alexander Fleming had identified the antibacterial properties of the *Penicillium* mold (see 1928), Australian biologist **Howard Walter Florey** (1898–1968) and British biochemists **Ernst Boris Chain** (1906–79) and **Norman**

Heatley (1911–2004) not only demonstrated that its antibiotic secretion, **penicillin**, could be used to cure infection (see panel, above), but that it could be produced and isolated in useful amounts as well. Medical trials began in January 1941, and techniques for the **mass production of the antibiotic** would be developed by the time World War II was at its height.

At the same time, German-born American biochemist **Fritz Albert Lipmann** (1899–1986) had been studying the **chemistry of metabolism** and made a

breakthrough in understanding the way living cells process their energy. In 1941, he reported that a phosphate-rich substance called **adenosine triphosphate (ATP)** was the chemical key to the process. While ATP had been discovered more than 10 years earlier, biologists were only now able to recognize its function. By burning calorific nutrients, such as carbohydrates and fats, living cells harness their energy in the phosphate bonds of a pool of ATP molecules. When energy is required—such as for growth or movement—ATP is broken down, unlocking the energy of its phosphate bonds to do the work (see panel, opposite).

While working at the Dow Chemical Company, American engineer **Ray McIntire** (1918–96) had been asked to develop an insulating material as part

“ I WAS A 28-YEAR-OLD KID AND I DIDN'T STOP TO RUMINATE ABOUT IT. ”

Glenn Theodore Seaborg, American scientist, on being part of the team that discovered plutonium, 1947

1940 Reginald Aldworth Daly publishes *Strength and Structure of the Earth*

February 27, 1940
Chemists Martin Kamen and Samuel Ruben discover the radioactive substance carbon-14

June 8, 1940 American scientists Edwin M. McMillan and Philip Abelson announce the discovery of neptunium

August 24, 1940 Howard W. Florey, Ernst B. Chain, and Norman Heatley show how purified penicillin can cure bacterial infection

December 14, 1940 American physicists Edwin M. McMillan and Glenn T. Seaborg synthesize a new element, later named plutonium

December 28, 1940
Ernst B. Chain and British biochemist Edward Abraham describe evidence of antibiotic resistance in bacteria

January 1941 Medical trials of penicillin begin

1941 Canadian astronomer Andrew McKellar makes estimates of deep space temperature

1941 Fritz Albert Lipmann describes the function of ATP

1941 Ray McIntire devises a way to make foamed polystyrene, later called styrofoam